Laboratory 6: Calculator [8 Bit]

1 INTRODUCTION

In this lab you will be creating an 8 bit calculator with basic addition, subtraction, multiply, and divide operations. The calculator will also use a memory to save all historical results as well as feed back the present output back into the ALU. You are encouraged to use packages for this lab and to start building up your own IP library. The technical requirements for this lab are listed below.

1. **Shall** employ pushbuttons for 'execute', and 'reset'.
    a. Might have to do a reset inversion in the top level due to pushbutton polarity.
2. **Shall** use the 8 right-most switches for the second input to the calculator.
3. The first input into the calculator **shall** come from the latest result stored in memory. On reset, the latest result **shall** be zero.
4. **Shall** implement an 8 bit wide memory that contains the ALU results. Could create a 4x8 bit memory as shown below. The number or rows in your memory dictates how many results can be stored.

address	reg description
00	Resets to zero
01	First result
10	Second result
11	Third result

5. **Shall** use the 2 left-most switches to select the desired operation
    a. [0:0] add [0:1] sub [1:0] multiply [1:1] divide
6. The present result **shall** be displayed on 3 seven segment LEDs in decimal form.
7. Calculator inputs **shall** both be 8 bit and the calculator output shall be 8 bit.
8. **Shall** operate with unsigned base 256 numbers. Ex. 0 - 1 = 255
9. All inputs **shall** be synchronized with the 50 MHz clock.
10. **Shall** display the present state via LEDs.

2 PRE-LAB [BLOCK DIAGRAM]

You shall create an order of operations list as well as a block diagram with a state transition diagram accurately mapping out your design. Also bring a printout of the block diagram to your lab class for an open discussion. Make sure to include proper synchronization, edge detection, and signal names in your diagram as well as a clear notation indicating the bit width of various busses and whether a block is a component or process. A state transition diagram as well as an order of operations list is also required.

3 HARDWARE

Target your design onto the DE1 SoC and receive a signoff. To save you time, I posted my compile.tcl file that lists the pin assignments that I used.

4 LAB UPLOAD => REQUIRED

In order to get full credit for the lab, please do the following.

- ☐ Delete the **work** folder that modelsim generates [if you did a simulation] and the **project** and **output_files** folders in the hw folder.
- ☐ Zip up the high level lab folder
- ☐ Post the .zip file to the below link
  https://u.pcloud.com/#/puplink?code=dJKXZDPnigsyLGuYYqVTbtzQo6XbnddX7

5 SIGNOFFS- LAB 6 NAME:_____

Category	Initials	Date	Points
Block Diagram			/20
Demonstration			/80
Final Grade			/100